Certified enclosures of the CO_2 uptake per unit alkalinity in seawater equilibrium chemistry

(a model computation; the value ≈ 0.8 is long known)

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AI-assistance disclosure and audit status. This preprint, its Python and Lean code and its claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was independently audited by a different AI agent instance (`AUDIT.md`, `ACCEPT_WITH_CORRECTIONS`), not by a human.** The audit's ten corrections are applied in this text (Section 10). No human audit and no Lean statement lock exist; both are owed (Section 13).

Abstract

For seawater in equilibrium with a fixed fugacity of CO_2, adding one mole of alkalinity (total alkalinity, TA) raises dissolved inorganic carbon (DIC) by $\eta = \partial\mathrm{DIC}/\partial\mathrm{TA}$ moles. We compute rigorous interval-arithmetic enclosures of η over the box $T \in [0, 30]\,^\circ\mathrm{C}$, $S \in [30, 38]$, $f\mathrm{CO}_2 \in [280, 560]\,\mu\mathrm{atm}$, $\mathrm{TA} = (S/35)\,\mathrm{TA}_{35}$, $\mathrm{TA}_{35} \in [2200, 2400]\ \mu\mathrm{mol\,kg^{-1}}$, in a carbonate–borate–water model on the total pH scale, with constants as printed in the Dickson–Sabine–Christian Guide [1]. With the constants exactly as printed, $\eta \in [0.7036, 0.9557]$ over the whole box (Tier [C], given the constants); the true range contains a value ≥ 0.94428 and a value ≤ 0.74271. An independent AI audit certifies the same interval and a tighter one, [0.7410, 0.9455]. Algebra (the closed form, $0 < \eta < 1$) is checked in Lean (Tier [A], statement lock owed). **The value $\eta \approx 0.8$ and its dependence on temperature, salinity and $p\mathrm{CO}_2$ are long known**; what is added is an enclosure with stated constants and a monotone closed form. **What this is NOT:** no tonnes of CO_2 removed, no climate effect, no statement on ecological safety, no deployment advocacy. It is equilibrium thermodynamics of a model: no gas-exchange kinetics, no $CaCO_3$ saturation, no mixing, no organic alkalinity, surface pressure only. Also reported: a dilute-water (soil) pH scan (Tier [X]), and a partial reproduction of the alkalinity-front retardation arithmetic of Jessen et al., for which the printed numbers are not all mutually consistent.

What this is NOT. Not a statement about tonnes of CO_2 removed, about any climate effect, about the ecological or biogeochemical safety of ocean alkalinity enhancement (OAE) or enhanced weathering (EW), or about whether anyone should deploy either. Not a new phenomenon: $\eta \approx 0.8$ is textbook-level. Not verified against the primary papers for the constants (Section 5). Part (b) (soil/dilute water) is Tier [X] only.

Markers. [A]: kernel-checked Lean, axioms within {`propext`, `Classical.choice`, `Quot.sound`}, controls caught, no statement lock yet. [L]: a cited source, taken as stated. [C]: executed code with the argument written down, not machine-checked. [X]: floating-point or sampled comparisons, supporting nothing by themselves. Identifiers `CK2-·-nnnn` refer to `ledger/ledger.json`.

1 Model, quantity and scope

Model. Carbonate, borate and water alkalinity on the total pH scale, with $h = [\mathrm{H}^+]$ ($\mathrm{mol\,kg^{-1}}$, total scale):

$$\mathrm{DIC} = C\Big(1 + \frac{K_1}{h} + \frac{K_1K_2}{h^2}\Big), \qquad \mathrm{TA} = C\Big(\frac{K_1}{h} + \frac{2K_1K_2}{h^2}\Big) + \frac{B_T K_B}{K_B + h} + \frac{K_W}{h} - h, \qquad C = K_0\, f\mathrm{CO}_2.$$

Sulfate, fluoride, phosphate and silicate are set to zero (their effect is quantified in Section 6). The total scale means $-h$ already stands for $-[\mathrm{H^+}]_F - [\mathrm{HSO_4^-}]$. Total boron $B_T = 0.000416\,S/35\ \mathrm{mol\,kg^{-1}}$. $f\mathrm{CO_2}$ is the independent variable (no fugacity coefficient). The TA definition is the carbonate–borate–water one just written; any other definition of TA (organic alkalinity, nutrient contributions) changes η (Section 6).

Quantity. At fixed $f\mathrm{CO_2}$ (C fixed), $\eta = (\partial\mathrm{DIC}/\partial h)/(\partial\mathrm{TA}/\partial h)$, which gives

$$\eta = \frac{C\,(K_1/h^2 + 2K_1K_2/h^3)}{C\,(K_1/h^2 + 4K_1K_2/h^3) + B_T K_B/(K_B+h)^2 + K_W/h^2 + 1} = \frac{h + 2K_2}{(h+4K_2) + h^3 R/(CK_1)},$$

$R = B_T K_B/(K_B+h)^2 + K_W/h^2 + 1$. Lean proves that the two forms agree (`eta_rewrite`) and $0 < \eta < 1$ for $h, K_1, K_2, C > 0$, $R \geq 0$. The producer's closed form was checked against a 50-digit numerical derivative of DIC(TA) at 50 random points (worst relative difference $1.3 \cdot 10^{-16}$) and, independently, by the auditor at three points including both extreme corners (agreement $5 \cdot 10^{-9}$, limited by the auditor's numerical-derivative step, not by the closed form).

Mean-value remark. A finite alkalinity addition $\Delta\mathrm{TA}$ has an efficiency equal to a mean of η over the TA segment; it lies in the enclosure if the whole segment lies in the box.

Scope. Equilibrium thermodynamics of a model. η is a model derivative, not a measurement-and-verification protocol.

2 Prior art

To our knowledge $\eta \approx 0.8$ at 25 °C, and its dependence on temperature, salinity, $p\mathrm{CO_2}$ and TA, is long known (for instance Zeebe and Wolf-Gladrow [3]; Renforth and Henderson 2017; the OAE efficiency literature), and the closed form is a ratio of buffer derivatives. None of these was re-read for this paper and no systematic literature search was made (neither by the producer nor by the auditor). The only addition attempted is a rigorous enclosure with stated constants and a monotone closed form.

3 Method

1. $\mathrm{TA}(h)$ is strictly decreasing in h, so h is bracketed rigorously (interval sign test of $\mathrm{TA}(h) - \mathrm{TA}$ at both ends, bisection tightening, `mpmath.iv`, 30 digits). The bracket is split into 24 pieces. On each piece where an interval test certifies $\mathrm{d}\eta/\mathrm{d}h > 0$, η is evaluated at the two ends only. Lean proves the algebraic identity $N'D - ND' = A\,(2Ah^2K_2 + Q)$ for hand-supplied N', D' and $Q \leq 0$ (`eta_deriv_identity`, `Q_nonpos`); **that N' and D' are the h-derivatives of N and D was checked by hand, by the producer and by the auditor, not in Lean.**
2. Global enclosure by branch-and-bound over $(T, S, f\mathrm{CO_2}, \mathrm{TA}_{35})$: the upper bound is the largest live-cell upper bound once every other cell is pruned against a rigorous point lower bound, and symmetrically for the lower bound. A cell is discarded only if its rigorous bound is beaten by a rigorous point bound (soundness re-read by the auditor). Constants per cell are enclosed by hulls over 24×6 sub-intervals. **Both U0 runs stopped at the evaluation cap (3000 evaluations, about 1300 live cells) and not at convergence, so the bounds are sound but not tight.**
3. Constants (Section 5): Weiss 1974 (K_0), Lueker et al. 2000 (K_1, K_2), Dickson 1990 (K_B), Millero 1995 (K_W), Uppstrom 1974 (B_T), as printed in the Guide [1].
4. Scenarios: U0, constants exactly as printed; U1 and U2, the pK values of K_0, K_1, K_2, K_B, K_W widened by $\pm 1\sigma$ and $\pm 2\sigma$ ($\sigma = 0.002, 0.0075, 0.015, 0.01, 0.01$, the values shipped in PyCO2SYS `uncertainty.pKs_OEDG18`, a secondary source). These are input intervals, not probability statements.

The preregistration (`prereg_K2.json`, sha256 `24b99955...50000`, verified by the auditor) was, by the producer's statement, written and hashed before any attempt; it was committed together with the results (commit `e0fbdba`), so git cannot confirm the order.

4 Results

4.1 (a) Seawater box, Tier [C] (given the pinned constants; not replayed in Lean)

scenario	producer's enclosure	independent audit
U0 (as printed)	[0.7036, 0.9557] (certificate [0.703565, 0.955707] rounded inward; rounded outward [0.7035, 0.9558])	certified, 223 cells, union [0.70390, 0.95563]; *tighter* [0.7410, 0.9455] certified (3039 cells); [0.7425, 0.9445] not certifiable (101 corner cells unresolved at depth 30, interval dependency)
U1 ($\pm1\sigma$ in pK)	[0.6826, 0.9618]	certified, 167 cells, union [0.68333, 0.96155]
U2 ($\pm2\sigma$ in pK)	[0.6616, 0.9676]	certified, 140 cells, union [0.66166, 0.96494]

All three producer enclosures are sound (every point of the box lies inside). The U2 row was re-run with the current code (Section 11) and is unchanged.

Sharp side (correction 1). The rigorous one-sided point bounds give: the true range contains a value ≥ 0.94428 and a value ≤ 0.74271 (the producer's point bounds are $0.9442832\ldots$ and $0.7427075\ldots$; the auditor's 50-digit corner values are 0.944283244 at (0 °C, S 30, 560 μatm, 2200) and 0.742707543 at (30 °C, 38, 280, 2400)). The producer's earlier rounded statements "≥ 0.9443" and "≤ 0.7427" do not follow from these bounds and are withdrawn. The U0 enclosure therefore overshoots the true range by at most 0.0115 above and 0.0392 below.

U1 and U2 (correction 2). The U1/U2 enclosures are wider than U0 by about 0.02/0.04 at the lower end, but this is mostly *interval overestimation*: the pK intervals are never subdivided. At the two extreme corners, over all 32 sign combinations of $\pm1\sigma$ ($\pm2\sigma$) on the five pK values, the attained values (point values, 50 digits, auditor) span [0.7374, 0.9468] (U1) and [0.7322, 0.9491] (U2), so the pK uncertainty moves the attained range by about 0.005 (0.010) at the bottom. This is Tier [X]: point values, not a certified 9-dimensional run; it was not proved that the extremes sit at corners, and no tight certified U1/U2 exists (a 9-D adaptive run with the pK offsets as extra dimensions did not finish in 10 min).

Reading. Per mole of alkalinity added, about 0.70 to 0.96 mol of DIC is taken up at equilibrium over this box in the model. On grids, η is larger when cold, at low salinity-normalised alkalinity and at high $p\mathrm{CO}_2$ (lower pH): finite differences on a 5^4 grid have uniform signs (η decreases with T, S, TA_{35} and increases with $f\mathrm{CO}_2$), and corners plus a 7^4 interior grid give the range [0.742708, 0.944283]. **These monotone trends are a numerical observation (Tier [X]), not certified** (correction 10).

4.2 (b) Dilute and soil water, Tier [X]

Same algebra without borate, with freshwater K_1, K_2 (Millero 1979 as transcribed in PyCO2SYS; the coefficients $290.9097/-14554.21/-45.0575$ and $207.6548/-11843.79/-33.6485$ were matched by the auditor to PyCO2SYS `p1atm.py` and give $\mathrm{pK}_1 = 6.351$, $\mathrm{pK}_2 = 10.329$ at 25 °C). For $f\mathrm{CO}_2 = 420\,\mu$atm and 0–30 °C: $\eta < 0.01$ below pH about 4.5 to 4.7, $\eta < 0.1$ below about 5.0 to 5.2, $\eta = 0.5$ at about 5.5 to 5.7 (outer pH intervals of width 0.13, `data/data/soil_result.json`); at $f\mathrm{CO}_2 = 10\,000\,\mu$atm every threshold is about 0.7 pH units lower. Reading: in this regime $\eta \approx [\mathrm{HCO}_3^-]/([\mathrm{HCO}_3^-] + h)$; the auditor's closed-form estimate $h^2 = CK_1$ puts $\eta = 0.5$ at pH 5.60 (25 °C) and 5.54 (0 °C), inside the producer's range. **Limitations (correction 7):** freshwater K_1, K_2 are not pinned to a primary source; the pK uncertainty ±0.02 is an assumption; activity corrections at soil-water ionic strength are ignored; and K_W is the seawater expression evaluated at $S = 0$ including the Guide's -0.015 total-scale shift, which does not apply in fresh water (K_W about 1.5% low, negligible below pH 6).

4.3 (c) Alkalinity-front retardation (arithmetic of Jessen et al.)

Jessen et al. [2] (preprint, "Subsoil acidity causes long delays in inorganic carbon sequestration by Enhanced Weathering") describe a downward-progressing front of alkalinity delayed by cation-exchange retardation $R = 1 + \Delta q/\Delta c$. With the delay $T(\mathrm{CEC}, L, q) = (L/q)(\varepsilon_w + k\,\mathrm{CEC})$, $k = 10\,\Delta\beta(1-\varepsilon)\rho/\Delta c$, equal to LR/v with $v = q/\varepsilon_w$ (Lean: `delay_via_R`), the delay is increasing in CEC, L, k and decreasing in q (Lean: `delay_mono_cec`, `delay_mono_L`, `delay_mono_k`, `delay_anti_q`; Tier [A], algebra only).

Reproduction (control P4; correction 6). The printed inputs (as read by the producer and re-read by the auditor from Supplementary Text 2) are $\Delta\beta = (0.55 - 0.2) + (0.12 - 0.03) = 0.44$, $\varepsilon = 0.3$, $\rho = 2.65$, $\varepsilon_w = 0.1$, $\Delta c = 0.80$ meq L^{-1}, unretarded velocity 5 m yr^{-1}, with CEC 0.56 (sum of exchangers) and 1.6 (direct). They give $\Delta q = 45.71$ and 130.59 meq L^{-1}, $R = 58.13$ and 164.24, $v = 8.60$ and 3.04 cm yr^{-1} and 11.6 to 32.8 yr m^{-1}. **R and v pass within printed rounding** (printed $R = 58$ and 165, $v = 8.7$ and 3.0 cm yr^{-1}). However the printed Δq values (45.4, 131.3) have ratio 0.3458, whereas Δq is proportional to CEC, which requires $0.56/1.6 = 0.35$; the printed R and v follow from the printed Δq (for example $1 + 45.4/0.8 = 57.75$ gives 8.66 cm yr^{-1}). The printed upper value "34 yr m^{-1}" is not reproduced (32.8 from the inputs, 33.3 even from the printed 3.0 cm yr^{-1}). **P4 is therefore a partial pass**, not a pass to the digit as preregistered. This is stated neutrally; the authors were not contacted, and the Jessen PHREEQC model and Dataverse files were not examined.

Table. Delay in yr m^{-1} over CEC and flux is in `data/data/delay_result.json` (for instance CEC $[1, 2]$, $q \in [0.5, 1]$ m yr^{-1}, $L = 1$ m: [10.3, 41.0]). A ±10% sensitivity on $\Delta\beta$ and Δc is Tier [X].

5 Constants and their provenance

The primary sources were not read by the producer or by the auditor: Lueker et al. 2000 (the auditor's attempt returned HTTP 403, and an eScholarship hit was an unrelated Scripps report), Weiss 1974 (an unofficial copy was found, the fetch failed), Dickson 1990 (no open copy found), Millero 1995, Uppstrom 1974 and Millero 1979 (not searched). The constants therefore rest on the Guide's transcription [1] (read; eqs. 30, 37, 40, 43, 62) and on PyCO2SYS 1.8.3.4 (GPL-3; used only, not redistributed).

- The Guide's five printed check values at $S = 35$, 25 °C ($\ln K_0 = -3.5617$, $\log_{10} K_1 = -5.8472$, $\log_{10} K_2 = -8.9660$, $\ln K_B = -19.7964$, $\ln K_W = -30.434$) are reproduced by the producer (5 of 5) and by the auditor, who typed the equations himself from the Guide text and obtained −3.561652, −5.8471529, −8.9659515, −19.796402, −30.433844.
- The producer's constants agree with PyCO2SYS's for K_0, K_1, K_2, K_B to 10^{-14}, an independent confirmation of the transcription (not of the Guide's own fidelity to the primary papers). K_W differs by −1.49% (the Guide subtracts 0.015 from the constant term to approximate the SWS-to-total conversion; PyCO2SYS converts exactly) and B_T by +0.07% (the Guide rounds Uppstrom's 0.0004157 to 0.000416).
- **No second CO2SYS-type package** (mocsy, seacarb) is available. The auditor's independent implementation is his own 50-digit `model.py` (constants typed from the Guide, own root solver and closed form). His PyCO2SYS cross-check at 202 points (seed 777, including both extreme corners): worst relative difference $4.4 \cdot 10^{-5}$ with sulfate = fluoride = 0, and $1.4 \cdot 10^{-10}$ when PyCO2SYS is fed the Guide's K_W and B_T.
- **Interval library.** `python-flint`/Arb is not installed for the auditor, so he too used `mpmath.iv` (30 digits): the code is independent, the interval library is not.

6 Omitted terms and caveats (correction 4)

- **Sulfate and fluoride** omitted: η changes by $+2 \cdot 10^{-6}$ to $+7.2 \cdot 10^{-5}$ over 202 sampled points (Tier [X], PyCO2SYS, auditor; the producer's "at most $6.8 \cdot 10^{-5}$" was a 100-sample maximum). With total-scale h the HSO_4^- term is already contained in $-h$; the real omissions are HF and the K_B/K_W scale conversions.

- **Phosphate** 2 **and silicate** 100 μ**mol kg**$^{-1}$ lower η by 0.0004 to 0.0015 (up to 0.0022 at 3/150 with sulfate and fluoride). Even then the result stays inside the U0 enclosure (minimum $0.7427 - 0.0022 = 0.7405 > 0.7036$).
- **Boron.** The Lee et al. 2010 alternative ($B_T = 0.0004326\,S/35$, about +4%) lowers η by 0.0004 to 0.0007 at the extreme corners ($-7.2 \cdot 10^{-4}$ and $-4.3 \cdot 10^{-4}$); Uppstrom's unrounded 0.0004157 changes it by $+1.3 \cdot 10^{-5}$ and $+7.8 \cdot 10^{-6}$; the K_W shift by $-4.6 \cdot 10^{-5}$ and $-2.1 \cdot 10^{-6}$.
- **Not bounded:** organic alkalinity (unbounded in principle for coastal waters) and pressure (surface, 1 atm only). Fugacity coefficient: none, because $f\mathrm{CO}_2$ is the input.
- The claim is **model-conditional on the total pH scale and on the TA definition stated in Section 1.**

7 Controls

control	result
P1 PyCO2SYS 1.8.3.4 (Lueker, total scale, sulfate = fluoride = 0), 500 random points, η from a central 1 μmol kg^{-1} difference	PyCO2SYS value inside the U0 enclosure of a small neighbourhood box ($T \pm 0.5$, $S \pm 0.25$, $f \pm 5$, $\mathrm{TA}_{35} \pm 5$) at 500/500; float agreement worst $4.1 \cdot 10^{-5}$. Point-box containment 0/500 is expected (the point box is 10^{-9} wide; the offset is $4 \cdot 10^{-5}$). The neighbourhood form was chosen, and P1b added, after seeing the offset.
P1b PyCO2SYS fed the Guide's K_W and B_T	100/100 inside the point enclosure ($+10^{-6}$), worst relative difference $6.7 \cdot 10^{-10}$: the $4 \cdot 10^{-5}$ offset is explained by those two inputs.
P2 closed form vs 50-digit numerical derivative	50 points, worst relative difference $1.3 \cdot 10^{-16}$.
P3 Guide printed check values	5/5 reproduced.
P4 Jessen arithmetic	partial pass (Section 4.3).
N1 $K_1 \leftrightarrow K_2$ swapped	100/100 PyCO2SYS values excluded by the corrupted model.
N2 wrong-sign borate term	100/100 excluded.
N3 $p\mathrm{CO}_2$ unit error (420 read as atm)	box guard rejects; forced solve returns no root.
N4 K_1 coefficient 3633.86 $\to$ 3733.86	$\log_{10} K_1 = -6.1826$, P3 value rejected.
N5 planted `sorry`/axiom in `Controls.lean`	compiles, rc 0; `#print axioms` shows `sorryAx`, resp. `CarbonateK2.ctl_cheat`: caught.

8 Lean (Tier [A]: algebra only)

Module `formal_cert/Programmes/CarbonateK2/Core.lean` (copy in `data/lean/`), compiled with `scripts/certified_numerics/lean_pinned.py` (Lean 4.34.1 with the pinned Mathlib), rc 0, 14 theorems, each `#print axioms` exactly {`propext`, `Classical.choice`, `Quot.sound`}. Statements: `eta_rewrite`; `eta_pos`, `eta_lt_one`; `eta_deriv_identity` and `Q_nonpos`; `delay_mono_cec`, `delay_mono_L`, `delay_anti_q`, `delay_mono_k`, `delay_via_R`; `jessen_R_sum`, `jessen_R_direct`, `jessen_yr_per_m_sum`, `jessen_yr_per_m_direct` (exact rational instances: $R \in (58, 59)$, $(164, 165)$; $\mathrm{yr\,m}^{-1} \in (11, 12)$, $(32, 33)$). `leanchecker Programmes.CarbonateK2.Core`: rc 0 (producer 68 s; auditor 64 s). Escape grep (`sorry`, `admit`, `native_decide`, `axiom`, `unsafe`, `implemented_by`, `extern`, `opaque`, `set_option`, `macro`, `elab`, `csimp`): no match (auditor).

Scope (correction 3). Lean proves algebra only, and nothing about the constants, the root h or the enclosures. In `eta_deriv_identity` the expressions N' and D' are supplied by hand (no `HasDerivAt`); that they are the derivatives of N and D was checked by hand by the producer and the auditor, **not in Lean**. The interval monotonicity test $2CK_1K_2/h^2 > |Q|$ is then a correct sufficient condition for $\mathrm{d}\eta/\mathrm{d}h > 0$ *given* that hand check. **No statement lock exists** (the repository has no `tools/statement_lock.py`, and no file outside K2 mentions `CarbonateK2`): Tier [A] pending lock. No person has certified that the statements mean what their gloss says.

Controls. `Controls.lean` compiles with rc 0 and is caught by the axiom gate (`ctl_sorry`: `sorryAx`; `ctl_axiom`: `CarbonateK2.ctl_cheat`); the auditor repeated this.

- `Core.lean`: `df5d68d91ec6076bbdb8dccec5d6c3f5750bcce91dcd3c7d49e629271f0db51f`
- `Controls.lean`: `1510f654e97b804a8a2b12e3a4d12eb16f071b7581b832ce24f097081d014bb8`

9 Independent AI audit

A different AI agent instance (`AUDIT.md`, 2026-10-31; verdict ACCEPT_WITH_CORRECTIONS; none of the producer's code used for the numerics; scripts in `data/audit_scripts/`, sources and small outputs only, no oleans) did the following.

- Re-derived the closed form and the derivatives $N' = -A(2h + 6K_2)$, $D' = -A(2h + 12K_2) - 2RB/(K_B + h) - 2R_W/h$ (notation of `AUDIT.md` §1), matching the hand-supplied expressions in Lean.
- Rigorous 4-D enclosure check with a different method: constants enclosed by a mean-value (centred) form with analytic T, S derivatives (checked against `mp.diff` to 10^{-30}), $\eta = 1/(1 + g)$ as a plain interval evaluation on 4 h-pieces with adaptive bisection, TA interval widened outward by 10^{-12} relative. Results in Section 4.
- Re-read the branch-and-bound pruning (sound; both U0 runs hit the cap of 3000 evaluations; `k2lib.eta_box` converts interval endpoints with `float()`, round-to-nearest, error at most 1 ulp, no reported digit affected).
- Constants: typed five equations from the Guide, matched the producer's, reproduced the 5 check values; PyCO2SYS cross-check at 202 points.
- Omitted terms (Section 6); Lean recompilation, axiom audit, `leanchecker` replay, controls caught; Jessen arithmetic; wording check (no tonnes, climate, safety or deployment sentence found); PyCO2SYS licence GPL-3 confirmed; all 28 file hashes in `result.json` match.

Not done by anyone: the primary-paper coefficients; a tight certified U1/U2; an independent interval library; a statement lock; the Jessen PHREEQC/Dataverse model; a literature novelty search; freshwater activity corrections.

10 Audit corrections applied

The audit required ten corrections to the producer's report (`data/RESULT.md`, the producer's *pre-audit* text, superseded by this paper). Numbering as in `AUDIT.md`:

1. **Sharp side.** "≥ 0.94428" and "≤ 0.74271"; overshoot 0.0115 above and 0.0392 below. U0 $[0.7036, 0.9557]$ kept (the producer's certificate rounded inward; certified by the auditor), plus the auditor's tighter $[0.7410, 0.9455]$ (Section 4).
2. **U1/U2.** The widening is mostly interval overestimation; attained shifts about 0.005 and 0.010 (Tier [X]).
3. **Lean scope.** Hand-written N', D'; algebra only; no statement lock (Section 8).
4. **Caveats.** Sulfate/fluoride $7.2 \cdot 10^{-5}$ (202 samples); nutrients; Lee boron; organic alkalinity and pressure not bounded (Section 6).
5. **Method.** Cap of 3000 evaluations with about 1300 live cells (Section 3).
6. **P4.** Partial pass; Δq ratio; "34 yr m^{-1}" not reproduced (Section 4.3).
7. **(b)** K_W at $S = 0$ with the seawater shift (Section 4.2).
8. **U2 seed.** See Section 11.
9. **Prereg ordering** by producer statement only (Section 3).
10. **Monotone trends** are a numerical observation, not certified (Section 4).

11 U2 and the earlier root-finder seed

The producer's report states that U2 was run with an earlier version of the float root seed in `k2lib.float_root` (an Anderson solver, later replaced by bisection) and that the archived U2 file hash belongs to that earlier code. The h bracket is verified by interval signs, so the seed does not affect rigor. Correction 8 asked for a re-run with the current code if cheap. It was: `run_seawater.py data U2` on cores $6, 7$ (the same 3000-evaluation cap, 467.5 s wall time against 693.8 s archived). The result is the identical enclosure [0.6616290773537888, 0.9675833420943311] with the same one-sided point bounds ($0.9323619\ldots$, $0.7888157\ldots$), the log differs from the archived `enc_U2.log` only in the final wall-time figure, and the JSON file differs from the archived one only in the `seconds` field, so its sha256 differs ($857f833b\ldots$ against $156ac9b4\ldots$). The archived file (whose hash is recorded in `result.json`) is left as it was; the re-run copies are in `data/rerun_U2/`. The auditor's independent check also certifies the U2 enclosure.

12 Claim ledger

`ledger/build_ledger.py` files 12 claims at Tiers [A] (`CK2-A-0001`, `-0002`), [L], [C] (the last [C] row being the headline) and [X]. Each carries a content-addressed evidence blob in `ledger/evidence/`; each Tier [A] claim also carries an `ai_audit` field. The gate field `audit` is left null: the audit was done by an AI agent, not a person, so the Elenchus gate keeps `LEDGER_UNAUDITED_TIER_A` open for each Tier [A] row. `ledger/run_gate.py` runs the unmodified gate; the verbatim output and exit code are in `ledger/gate_output.txt`.

13 Limitations and what remains unverified

- **Model only; no real-world claim.** No tonnes, climate effect, ecological safety or deployment advice. Equilibrium thermodynamics: no gas-exchange kinetics, no $CaCO_3$ precipitation or saturation state, no mixing.
- **Constants not verified against primary papers**; they rest on the Guide transcription plus PyCO2SYS (Section 5). The kill criterion of the preregistration (constants not pinned to primary sources ⇒ Tier [X]) was applied per part: (a) is reported as Tier [C] *with this caveat*, (b) as Tier [X].
- **AI audit only; human audit and Lean statement lock owed.** No second CO2SYS package; interval library not independent.
- **U0 is loose** (branch-and-bound stopped at the cap); **U1/U2 are loose** (pK intervals not subdivided); no tight certified U1/U2.
- **Monotone trends** and the sharp-side corner values are Tier [X]/point bounds.
- **Literature novelty** not searched; **Jessen** PHREEQC model and Dataverse files not examined (licence not verified).
- **Deviations from the preregistration.** P4 tolerance relaxed to printed rounding (and then found a partial pass); P1 operationalised as a neighbourhood box and P1b added after seeing the $4 \cdot 10^{-5}$ offset; the time box (≈ 4 agent-hours) was exceeded because the branch-and-bound was redesigned twice (the first grid enclosures were far too loose, $\eta \in [0.69, 0.96]$ with 12% TA spread per cell); the `grid` entry in `seawater_result.json` is a leftover of the abandoned design and is not used.
- **Outcome.** PARTIAL: (a) certified given the pinned constants, (b) Tier [X], (c) algebra in Lean and arithmetic reproduced with an inconsistency in the source.

14 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`, HEAD `c574b58` (full: `c574b58aba67a9041406570e20d557a6c45a25a6`) at writing; the K2 outputs, the Lean files and this paper are uncommitted working-tree files; nothing was pushed. Copies of the inputs are in `data/` (`RESULT.md`, `result.json`, `summary.json`, `prereg_K2.json`, `AUDIT.md`, `code/`, `data/`, `logs/`, `lean/`, `audit_scripts/`). The producer

and the auditor ran on cores 6, 7 only. The bundler keeps only listed file suffixes, so `.log` and `.sha256` files are also stored as byte-identical `.txt` copies in `data/`; compiled Lean objects and third-party PDFs are not shipped. PyCO2SYS (GPL-3) is imported by some scripts but not redistributed.

```
cd results/proposal4/2026-10-27/K2
taskset -c 6,7 .venv/bin/python -I code/run_seawater.py data U0   # also U1, U2
taskset -c 6,7 .venv/bin/python -I code/run_seawater.py data controls
taskset -c 6,7 .venv/bin/python -I code/run_soil_delay.py data
.venv/bin/python -I code/controls_P2_P3.py
python -I scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/CarbonateK2/Core.lean
python -I papers/p4_ocean_alkalinity_uptake/ledger/build_ledger.py
python -I papers/p4_ocean_alkalinity_uptake/ledger/run_gate.py
```

References

[1] A. G. Dickson, C. L. Sabine, J. R. Christian (eds.), *Guide to Best Practices for Ocean CO_2 Measurements*, PICES Special Publication 3 (2007), Ch. 5 sec. 7 (eqs. 30, 37, 40, 43, 62 and the printed check values) ([L]; the PDF is not redistributed).

[2] S. Jessen, R. Jakobsen, M. Looms, P. Ambus, D. Postma, *Subsoil acidity causes long delays in inorganic carbon sequestration by Enhanced Weathering*, preprint, Supplementary Text 2 ([L]; recorded in the preregistration as arXiv 2607.01835, not re-verified from the PDF text; the PDF is not redistributed).

[3] R. E. Zeebe, D. Wolf-Gladrow, *CO_2 in Seawater: Equilibrium, Kinetics, Isotopes*, Elsevier (2001); P. Renforth, G. Henderson, Rev. Geophys. 55 (2017): named as prior art only, not re-read.

[4] Primary sources of the constants, *not read*: Weiss 1974; Lueker, Dickson, Keeling 2000; Dickson 1990; Millero 1995; Uppstrom 1974; Millero 1979; Lee et al. 2010 (boron alternative, via the auditor's variant only).